

Notation reference

Thomas Serre · Computational Methods for Mind, Brain & Behavior · Fall 2026 · one notation for the whole semester

The rules (read these first)

Type style. Scalars are lowercase italic (a, x, y); vectors are **bold lowercase** and are column vectors ($\mathbf{x}, \mathbf{h}, \mathbf{b}$); matrices and tensors are **bold uppercase** ($\mathbf{W}, \mathbf{A}^{[l]}$).

A layer has no transpose. Weights are stored output-rows-first, so $\mathbf{W}^{[l]} \in \mathbb{R}^{D_l \times D_{l-1}}$ (rows index the units of layer l , columns the units feeding in). Therefore

$$\mathbf{z}^{[l]} = \mathbf{W}^{[l]} \mathbf{a}^{[l-1]} + \mathbf{b}^{[l]}, \quad \mathbf{a}^{[l]} = g^{[l]}(\mathbf{z}^{[l]}).$$

This matches PyTorch `nn.Linear(in, out)` whose weight has shape `(out, in)`. The transpose appears *only* for a single neuron's dot product, $z = \mathbf{w}^\top \mathbf{x} + b$, because $\mathbf{w}$ is then a column vector.

Index discipline (the part that usually drifts). Position encodes meaning, so the same letter never means two things:

Index	Notation	Example	Range
sample / example	superscript (i)	$\mathbf{x}^{(i)}$	$i = 1, \dots, N$
layer	superscript $[l]$	$\mathbf{a}^{[l]}$	$l = 0, \dots, L$
sequence time (RNN)	superscript $\langle t \rangle$	$\mathbf{h}^{\langle t \rangle}$	$t = 1, \dots, T$
diffusion step	subscript t	$\mathbf{x}_t$	$t = 0, \dots, T$
unit / component	subscript j	a_j	–
repetition (trial) of sample i	superscript (i, r)	$\mathbf{x}^{(i, r)}$	$r = 1, \dots, R$
matrix entry	subscripts ij , lowercase	$x_{ij} = x_j^{(i)}$	row i : sample; col j : feature
RDM entry	subscripts ik , lowercase	$d_{ik} = d(\mathbf{x}^{(i)}, \mathbf{x}^{(k)})$	$\mathbf{D} \in \mathbb{R}^{N \times N}$
dissimilarity rules	lowercase subscript	$d_{\text{euc}}, d_{\text{cos}}, d_{\text{corr}}$	$1 - \cos \phi, 1 - r$
channel (CNN)	subscript c	$A_{:, :, c}$	$c = 1, \dots, C_l$
attention head	superscript (h)	$\mathbf{Q}^{(h)}$	–

Sequence time is a *superscript* $\langle t \rangle$; diffusion time is a *subscript* t . They never co-occur, and the position tells them apart.

Cross weights use destination-from-source subscripts: $\mathbf{W}_{hx}$ maps $\mathbf{x} \rightarrow \mathbf{h}$, $\mathbf{W}_{hh}$ maps $\mathbf{h} \rightarrow \mathbf{h}$, $\mathbf{W}_{yh}$ maps $\mathbf{h} \rightarrow \hat{\mathbf{y}}$.

Master symbol table

Symbol	Name	Notes
<i>Core (everywhere)</i>		
$\mathbf{x}, \mathbf{y}, \hat{\mathbf{y}}$	input, target, prediction	bold lowercase column vectors
$\mathbf{z}, \mathbf{a}, \mathbf{h}$	pre-activation, activation, hidden	$\mathbf{a} = g(\mathbf{z})$; $\mathbf{h}$ for recurrent/latent state
$\mathbf{W}, \mathbf{w}, \mathbf{b}$	weight matrix, weight vector, bias	$\mathbf{w}$ = one unit's incoming weights (a row of $\mathbf{W}$)
$g(\cdot), \sigma, \text{ReLU}, \text{softmax}$	activations	g generic; σ reserved for sigmoid only
θ, η	all parameters, learning rate	θ collects every trainable weight/bias
$\mathcal{L}, \ell, \delta$	total loss, per-example loss, error	$\delta = \mathbf{y} - \hat{\mathbf{y}}$ (also the backprop delta)
$\mathcal{D}, \mathcal{B}, \mathbb{E}$	dataset, minibatch, expectation	$\mathcal{D} = \{(\mathbf{x}^{(i)}, \mathbf{y}^{(i)})\}_{i=1}^N$
N, L, T, K	counts	examples, layers, seq. length, classes
<i>Theme A – single neuron</i>		
$V, V_{\text{rest}}, V_{\text{th}}$	membrane / resting / threshold potential	threshold is V_{th} , not θ
I, τ_m, r	input current, time constant, firing rate	for spiking / rate models
$z = \mathbf{w}^\top \mathbf{x} + b$	rate-neuron pre-activation	the one canonical transpose ($\mathbf{w}$ a column vector)
<i>Theme B – populations</i>		
$s, f_i(s), s_i^{\text{pref}}$	stimulus, tuning curve, preferred stim.	population-code basics
$\mathbf{r}, \hat{s}, I_F(s)$	population rate, decoded estimate, Fisher info	encoding/decoding
$\mathbf{W}^{[l]}, \mathbf{b}^{[l]}, \mathbf{z}^{[l]}, \mathbf{a}^{[l]}$	MLP layer quantities	$\mathbf{a}^{[0]} = \mathbf{x}, \mathbf{a}^{[L]} = \hat{\mathbf{y}}$
$\mathbf{W}_{hh}, \mathbf{W}_{hx}, \mathbf{W}_{yh}$	recurrent / input / output weights	dest-from-source subscripts
$\mathbf{h}^{\langle t \rangle}, \mathbf{x}^{\langle t \rangle}, \hat{\mathbf{y}}^{\langle t \rangle}$	RNN state / input / output at time t	superscript $\langle t \rangle$

Symbol	Name	Notes
$A^{[l]}, K^{[l]}, C_l$	feature map, kernel bank, channels	$A^{[l]} \in \mathbb{R}^{H_l \times W_l^{\text{sp}} \times C_l}$
<i>Theme C – modern deep learning</i>		
X, Q, K, V	token matrix, queries, keys, values	rows = tokens (the row-stacking exception)
W_Q, W_K, W_V, W_O	projection matrices	$Q = XW_Q$, etc.
$d_{\text{model}}, d_k, d_v, n_h$	model / key / value dims, # heads	
x_0, x_t, ϵ	clean data, noised sample, noise	diffusion step = subscript t
$\epsilon_\theta, \beta_t, \bar{\alpha}_t$	noise net, schedule, cumulative $\bar{\alpha}$	$\alpha_t = 1 - \beta_t, \bar{\alpha}_t = \prod_{s \leq t} \alpha_s$

The same model written each way

Single neuron (rate / perceptron)

$$z = \mathbf{w}^\top \mathbf{x} + b, \quad a = g(z), \quad \hat{\mathbf{y}} \text{ uses } \sigma \text{ for probabilities.}$$

MLP (deep feedforward) – the template every later model reuses

$$\mathbf{a}^{[0]} = \mathbf{x}, \quad \mathbf{z}^{[l]} = \mathbf{W}^{[l]} \mathbf{a}^{[l-1]} + \mathbf{b}^{[l]}, \quad \mathbf{a}^{[l]} = g^{[l]}(\mathbf{z}^{[l]}), \quad \hat{\mathbf{y}} = \mathbf{a}^{[L]}.$$

Recurrent network

$$\mathbf{h}^{(t)} = g(\mathbf{W}_{hh} \mathbf{h}^{(t-1)} + \mathbf{W}_{hx} \mathbf{x}^{(t)} + \mathbf{b}_h), \quad \hat{\mathbf{y}}^{(t)} = g(\mathbf{W}_{yh} \mathbf{h}^{(t)} + \mathbf{b}_y).$$

Convolutional layer

$$\mathbf{A}^{[l]} = g(\mathbf{K}^{[l]} * \mathbf{A}^{[l-1]} + \mathbf{b}^{[l]}), \quad \mathbf{A}^{[l]} \in \mathbb{R}^{H_l \times W_l^{\text{sp}} \times C_l}.$$

Self-attention (single head)

$$\mathbf{Q} = \mathbf{XW}_Q, \quad \mathbf{K} = \mathbf{XW}_K, \quad \mathbf{V} = \mathbf{XW}_V, \quad \text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{QK}^\top}{\sqrt{d_k}}\right)\mathbf{V}.$$

Diffusion (DDPM)

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}), \quad \mathcal{L} = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2].$$

Migration notes for existing decks. The Week 2-3 slides already use $\mathbf{x}, \mathbf{w}, b, \sigma, \eta, \hat{y}, \delta$ – keep them. The one change to standardize: write multi-layer weights as uppercase $\mathbf{W}^{[l]}$ (some slides currently use lowercase w_1, w_2 for layer matrices).